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# Course: Bioinformatics – ENBC 455
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Problem 1

```
with open('hw1_1.txt', 'r') as file:
    string = file.read()
    A = 0
    C = 0
    G = 0
    T = 0
    for char in string:
        if char == 'A':
            A += 1
        elif char == 'C':
            C += 1
        elif char == 'G':
            G += 1
        elif char == 'T':
            T += 1
    print(A, C, G, T)
    print()
```

Problem 2

```
with open('hw1_2.txt', 'r') as file:
    string = file.read()
    for char in string:
        if char == 'T':
            char = 'U'
        print(char, end='')
    print()
```

Problem 3

```
with open('hw1_3.txt', 'r') as file:
    string = file.read()
    reversed_string = string[::-1]
    for char in reversed_string:
        if char == 'A':
            print('T', end="")
        elif char == 'C':
            print('G', end="")
        elif char == 'G':
            print('C', end="")
        elif char == 'T':
            print('A', end="")
    print("\n")
```

Problem 4

```
from Bio.Seq import Seq
with open('hw1_4.txt', 'r') as file:
    seq = Seq(file.read().replace('\n', ' ').strip())
    print(seq.translate())
```